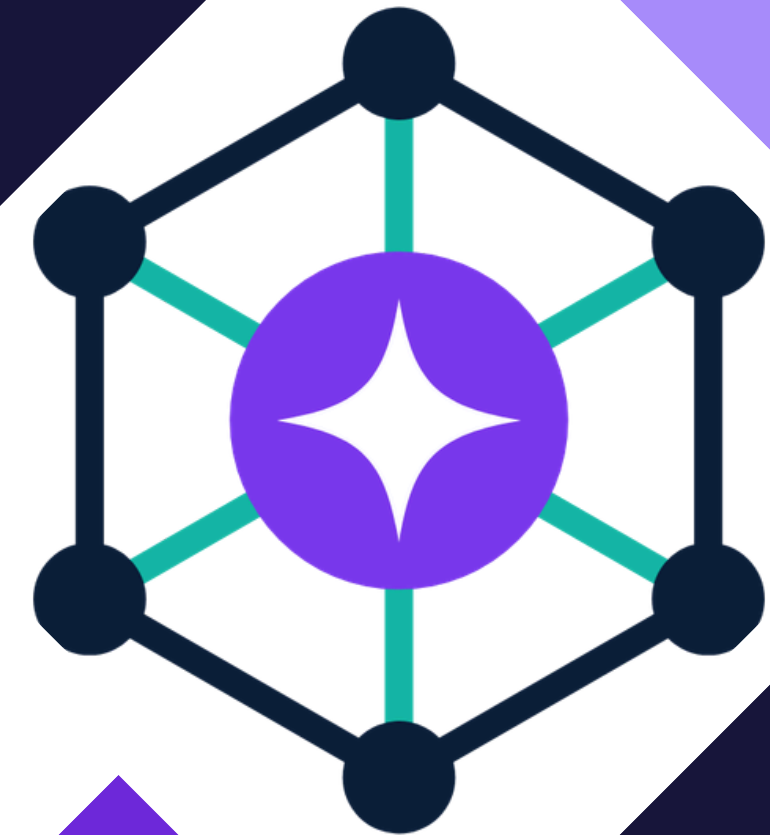


Self Network Deployer

AI-DRIVEN AUTOMATED DEPLOYMENT OF 5G TEST ENVIRONMENTS

M1: Inception



TEAM



Filipe Nogueira



Francisco Santos



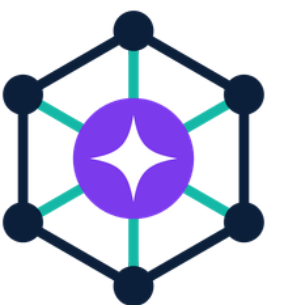
João Moraes



Mario Santos



Samuel Ramos



Research Group



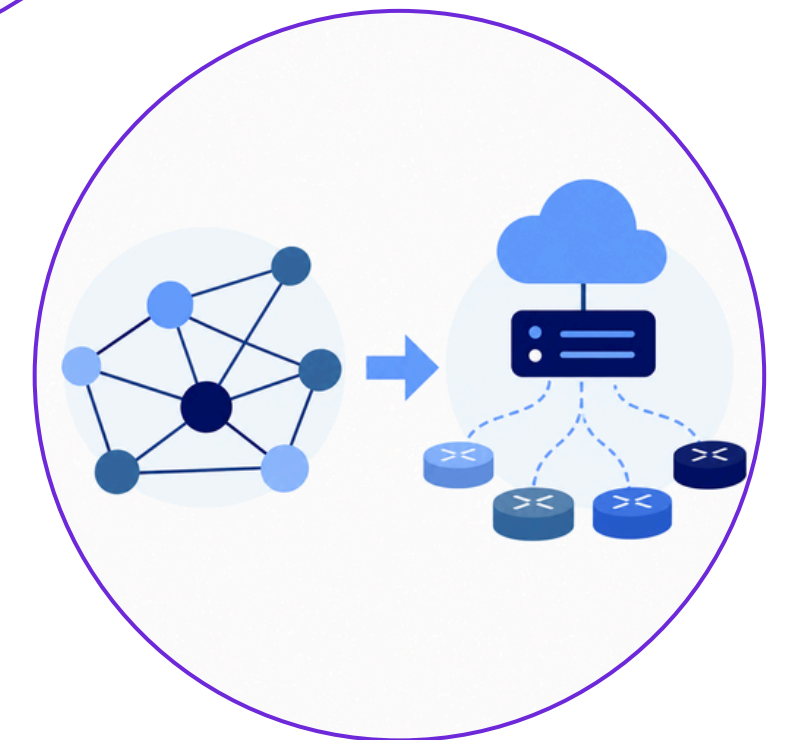
Supervision Team



Daniel Corujo

Context

- **EVOLUTION TOWARDS DYNAMIC AND PROGRAMMABLE NETWORKS**
- **INTEGRATION OF COMMUNICATION AND COMPUTING RESOURCES**
- **AI AND SOFTWAREIZATION**



Problems

- **Time-Consuming Manual Configuration**
- **Increasing Network Complexity**
- **Tool-Specific Knowledge**
- **Slow Network Adaptation**



Related Work

- A 5G core has many network functions that must be configured consistently
- Intent-Based Networking: the operator says what the network should do, not how [1]
- LLMs need documentation (RAG) and verification to be reliable [2][3][4][5]
- Generate → verify → deploy agents reached 94.4% configuration success [6]
- Takeaway: LLMs become much more reliable when grounded in documentation and verified [2][6]
- SND applies this approach to Open5GS [16]



Related Work

Other Works (Nephio, Cisco NSO, ONAP)



- Manual YAML / YANG / TOSCA coding
- No Natural Language understanding
- Static, manual error handling
- Heavy, complex infrastructure setup

Our Project (SND)



- Natural Language Intent interface
- Cognitive RAG documentation parsing
- AI Self-Healing & 3GPP Guardrails
- Sandboxed 5G closed-loop testbed

Goals

1. AI-ASSISTED NETWORK DEPLOYMENT

Develop an AI agent capable of assisting in the creation, configuration and deployment of network and computing infrastructures.



2. NATURAL LANGUAGE INTERACTION

Allow users to describe the desired infrastructure in natural language and translate their requirements into configurations and actions.

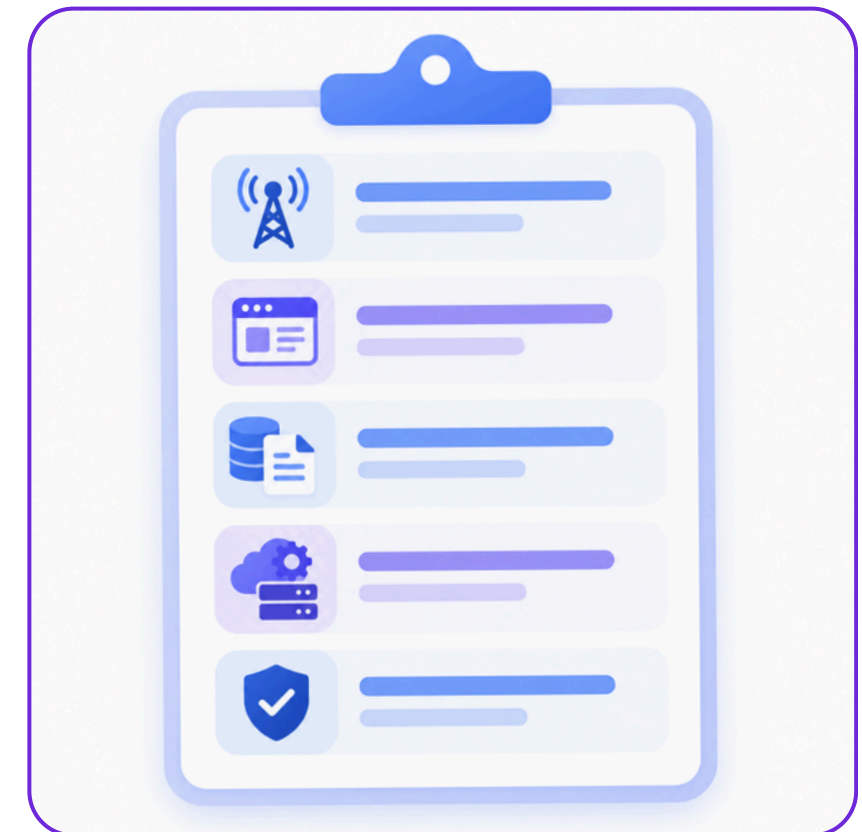


3. KNOWLEDGE-BASED AUTOMATION

Use technical documentation to configure and deploy different systems while validating generated configurations and correcting errors.

Tasks

- 1. 5G Environment & Infrastructure Analysis**
- 2. AI Knowledge Base & RAG Development**
- 3. API & User Interface Development**
- 4. Configuration Generation & Automated Deployment**
- 5. Deployment Validation & Feedback**



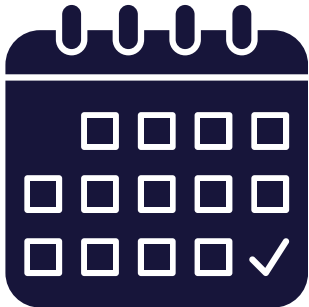
Expected Results

- **Functional SND Agent** – Natural language network deployment
- **Knowledge Base & RAG** – Knowledge retrieved from technical documentation
- **Deployment Planning** – Generation of deployment plans and configurations
- **Automated Deployment** – Configuration and deployment of 5G components
- **Validation & Error Handling** – Detection and correction of deployment errors
- **User Interface** – Deployment status, results and errors

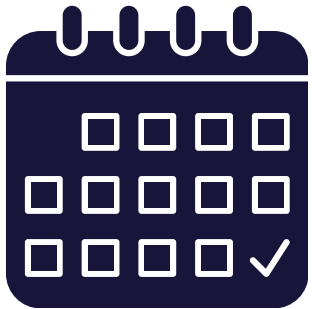


Communication plan





Milestone / Phase	Date	Task List
Inception	22/09/2026 → 29/09/2026	Project Website
	22/09/2026 → 29/09/2026	GitHub Organization / Repository <i>(pending approval)</i>
	22/09/2026 → 29/09/2026	Project Calendar
	22/09/2026 → 29/09/2026	Communication Plan
	22/09/2026 → 29/09/2026	Context & Problem
	22/09/2026 → 29/09/2026	Goals & Task Planning
	22/09/2026 → 29/09/2026	Expected Results
	22/09/2026 → 29/09/2026	Related Work / State of the Art
M1	29/09/2026	Presentation
Elaboration	30/09/2026 → 13/10/2026	Requirements Assessment
	30/09/2026 → 13/10/2026	User Stories
	30/09/2026 → 13/10/2026	Use Cases
	30/09/2026 → 13/10/2026	System Architecture
	30/09/2026 → 13/10/2026	Data Model Definition
	30/09/2026 → 13/10/2026	Architecture Validation
M2	13/10/2026 → 20/10/2026	Lifecycle Architecture Presentation / Validation



Construction	14/10/2026 → 03/11/2026	Accessibility Requirements & User Interface Design
	14/10/2026 → 03/11/2026	Interface Prototype & Usability/Accessibility Evaluation
MS3	03/11/2026 → 10/11/2026	Digital Accessibility and Usability
Construction	04/11/2026 → 24/11/2026	Existing LLM Integration
	04/11/2026 → 24/11/2026	Basic Open5GS Knowledge Base & RAG
	04/11/2026 → 24/11/2026	Basic 5G Network Data Model
	04/11/2026 → 24/11/2026	User Intent Processing
	25/11/2026 → 07/12/2026	Basic 5G Deployment Planner
	25/11/2026 → 07/12/2026	Open5GS Configuration Generation
	08/12/2026 → 14/12/2026	Basic Configuration Validation
	08/12/2026 → 14/12/2026	Basic Feedback & Error Handling
	08/12/2026 → 14/12/2026	Minimal API & User Interface
	08/12/2026 → 14/12/2026	Basic Logging
	08/12/2026 → 15/12/2026	MVP Integration & Testing
MS4	15/12/2026 → 16/12/2026	MVP Presentation with Supervisors & Peer Evaluation

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- [15]** OASIS, "TOSCA Version 2.0," OASIS Standard, Jul. 2025.
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Questions?



For more information:



<https://mariosantosp.github.io/SND-microsite/>